PROJECT-BASED LEARNING

PHASE-II EVALUATION

Smart Driver Drowsiness & Safety System

Team ID: DSCPP-III-2026-T234

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T234
Semester: 3rd
Section: H, E, ML4, ML3
Domain: *Computer Vision & Safety*

Team Members

1. *Ridvik Garg – 2028039*
2. *Prabhav Gulyani – 2027975*
3. *Nandini Gupta – 2028777*
4. *Antriksh Gehlot – 2028648*

Mentor

Mr. Kamal Kumar Gola

Interactions completed: 04

Brief Discussion: Introduce the team, project domain, mentor, and responsibilities of the team members.

Phase-I Recap & Feedback Incorporated

Phase-I Status

Proposed a webcam system to track driver eyes and head position to prevent accidents from sleepy drivers.

Feedback Received

- *Use proper DSA instead of standard vectors*
- *Split project into smaller testable modules*
- *Avoid false beeps on normal eye blinks*

Changes Incorporated

- *Added Queue, Stack, and Linked List classes*
- *Divided code into Vision, DSA, and DB modules*
- *Checked longer eye closure before alerting*

Brief Discussion: Explain the major feedback received during Phase-I and clearly demonstrate how it has been

Problem Statement & Finalized Objectives

Problem Statement

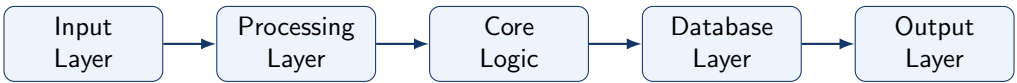
Prevent accidents caused by sleepy drivers by tracking prolonged eye closure via webcam and giving step-by-step alerts.

Finalized Objectives

1. *Capture live webcam video with OpenCV to check eye closure and head position.*
2. *Use a Queue for incoming events, a Stack for recent alerts, and a Linked List for history.*
3. *Play step-by-step sound alerts, lower car music volume, and send emergency SMS.*
4. *Save all past safety events and logs inside a local SQLite database.*

Brief Discussion: Avoid repeating the complete Phase-I presentation. Focus on the finalized problem, scope and measurable objectives.

Updated System Architecture / Workflow



Major Components

- *OpenCV Camera & Face Tracking*
- *C++ Queue, Stack & Linked List Logic*
- *SQLite Database & Alert Triggers*

Changes Since Phase-I

- *Added custom DSA classes for events*
- *Added music mute and SMS features*
- *Split into separate testable modules*

Brief Discussion: Present the updated architecture and explain how individual modules communicate with each other.

Module Design & Functional Requirements

Major Modules

1. *Webcam & OpenCV Vision Module*
2. *C++ DSA Event Handler Module*
3. *SQLite Database Storage Module*
4. *Sound, Music Mute & SMS Module*

Functional Requirements

- *Track eyes and verify prolonged closure*
- *Push events to Queue and alerts to Stack*
- *Save event history in Linked List & DB*
- *Trigger beep, loud alarm, mute, and SMS*

Module Dependency

OpenCV spots eye closure and sends events to the C++ Queue, which updates the Stack, triggers alerts, and saves logs to SQLite.

Technology Stack & Development Methodology

Technology Stack

- Programming: *C++*
- Database: *SQLite*
- Framework: *OpenCV*
- Development Tool: *VS Code & GCC*
- Version Control: *Git / GitHub*

Development Methodology

1. Requirement Analysis
2. System Design
3. Module Development
4. Integration
5. Testing & Validation

Brief Discussion: Explain why the selected technologies are appropriate and how the team is following the development methodology.

Implementation of PBL Course Concepts

Course	Concepts Actually Implemented	Project Module
Data Structures in C	*Queue (event order), Stack (recent alerts), Linked List (history)*	*DSA Event Module*
OOPs with C++	*Custom Classes, Objects, Encapsulation, Header Files*	*Core C++ Backend*

Important: Mention only those concepts that have been **actually implemented** in the project.

Brief Discussion: Students should explain where and how each course concept is technically implemented.

Development & Implementation Progress

Module	Status	Work Completed
C++ & DB Setup	Completed	*Set up C++ repo, SQLite tables, and tested insert/read queries*
Webcam Tracking	Completed	*Tested webcam input and verified basic face tracking with OpenCV*
DSA & EAR Logic	In Progress	*Debugging Stack/Linked List classes and adjusting low-light EAR*
Alerts & Merging	Pending	*Sound alerts, music mute script, SMS API, and full module merge*

Overall Project Completion: *55%*

Brief Discussion: Clearly differentiate between completed, in-progress and pending development. Do not present proposed work as completed work.

Working Prototype / Demonstration

Implemented Features

- *C++ build compiles cleanly*
- *Webcam live video capture*
- *Basic face & eye tracking*
- *SQLite insert & read queries*

Prototype / Output

Webcam face tracking & SQLite database test output ready

Brief Discussion: Demonstrate the actual working implementation. Students should preferably provide a live demonstration in addition to screenshots wherever applicable.

Testing, Results & Validation

Test / Module	Expected Result	Observed Result	Status
C++ & Webcam	*Build files compile and camera runs*	*Compiles fine and captures video stably*	Pass
DB Initialization	*Insert and fetch safety records*	*Database is ready and queries work*	Pass
DSA & Vision	*No memory leaks; track in dark rooms*	*Fixing leaks and tuning low-light EAR*	*In Progress*

Results / Validation

C++ build, webcam feed, and SQLite queries passed testing; eye tracking works in good light and is being tuned for dark rooms.

Brief Discussion: Results should be supported by actual testing or experimental evidence rather than assumptions.

GitHub Repository & Development Activity

Repository Information

- GitHub Link: *github.com/ridvik74/Smart-Driver-Drowsiness-Safety-System/tree/main/DSA*
- README.md: *Updated*
- Contributors: *All 4 Members Added*
- Repository Status: *Public*
- Branches: *main*

Development Activity

- *Custom DSA headers uploaded*
- *Camera processing scripts added*
- *Individual commits from own accounts*
- *SQLite database scripts uploaded*
- *Alert generation logic in progress*

Requirement: All team members must be contributors and must **commit individually using their own GitHub accounts**.

Brief Discussion: Show repository activity, README, contributor list and commit history during the presentation.

Individual Contribution

Member	Role	Technical Contribution	GitHub Commits
Ridvik Garg	*Team Lead & DSA*	*Implemented Queue, Stack, Linked List, Searching, Sorting, and main program integration.*	*Active*
Prabhav Gulyani	*Camera & Event Processing*	*Worked on OpenCV face/eye detection and safety-event processing.*	*Active*
Nandini Gupta	*Vision & Database*	*Tested OpenCV face tracking; developed SQLite database operations and queries.*	*Active*
Antriksh Gehlot	*Database & Alerts*	*Worked on database/alert-related modules and integration testing.*	*Active*

Challenges, Remaining Work & Phase-III Roadmap

Challenges Faced

- *Dark lighting and spectacles affect OpenCV*
- *Bugs while replacing vectors with custom DSA*
- *Crashes and CPU lag when linking modules*

Solutions / Learning

- *Testing different EAR values for low light*
- *Module-by-module testing to fix memory leaks*

Phase-III Roadmap

1. *Link eye-tracking with main event Queue*
2. *Finish debugging Stack and Linked List*
3. *Add sound alerts and music mute script*
4. *Set up SMS API for emergency texts*
5. *Merge all modules and do full testing*

Expected Outcomes & References

Expected Final Outcomes

- *Working driver safety prototype*
- *Real-time Queue, Stack & Linked List*
- *Step-by-step alarm and music mute*
- *Emergency SMS and SQLite logs*
- *Complete source code on GitHub*

References

1. *Real-Time Eye Blink Detection (EAR Paper)*
2. *OpenCV Face Landmark Documentation*
3. *Data Structures in C/C++ Textbook*
4. *Official OpenCV & SQLite3 Docs*
5. *GitHub Repository: Smart-Driver-Safety*

Thank You